

I first registered without the first name filled in and it cleared the form and did not register me. I did the same for the last name and it cleared the form and again did not register me. I registered without a username and got the same result. I registered with an existing username and got the same result. I registered without an email and got the same result. I registered without the @ symbol in my email and got the same result. I registered without a period in the email and got the same result. I registered without a password and got the same result. I registered with a password that was shorter than eight characters and got the same result. After testing all those fields, I entered in all the proper information and it logged me in and directed me to the My Suggestions page. I also checked the database that the user was actually saved.

It accomplishes this by first taking the username and comparing it against the other usernames in the user table. I then checked to make sure that the first and last name fields had information in the field. I would then check that the email had the @ sign and the period. I would also check to make sure the password was eight characters long. If it did not already exist, I would take all the input information and

create a new user, I hashed the password before saving it to the table. From there I logged in as the user by saving the username by passing it to a variable that when called, passes the logged in user. It then redirected them to the My Suggestions page.

User is able to logout.

I logged in as an existing user. I then went to the homepage and hit the log out button and was logged out. I logged in again and went to the My Suggestions page and hit the log out button and was logged out. I logged in again and went to the Their Suggestions page and hit the logout button and it logged me out.

It does this by a button click. When the log out button is clicked it forgets the username in that session, it does not delete it. It then has you on the website as not logged in.

User is able to click on the different categories.

I clicked through all the different categories and sub categories and made sure they went to the correct sub categories. I double checked with the database that the subcategories are the correct ones being shown.

It does this by collecting the ID number of the category clicked. It passes it to python. In the python it takes that ID and gathers all the categories that parent ID equals that ID. It then passes back to JavaScript and outputs them.

User is able to use the thumbs up and thumbs down to update the rating for the categories.

I clicked the thumbs up and down buttons and made sure that within each category it only allowed either a thumbs up or down. An example is that if someone thumbs up food then goes back to thumbs down food, the thumbs up would be unhighlighted. Also, that each category associated with thumbs up and down work independently. I made sure to check this in all the sub categories as well.

I did this by using JavaScript that is passed to the category ID number so that it knows which category it is being associated with. It will then select the one you clicked and deselect the one that was previously clicked, if that applies.

User is able to use the previous button to go back to the previous page.

I have the previous button in three places. One in the different categories. I clicked on the button and it took me back to the previous categories. The second place is when the user clicks on my suggestions. I clicked on that button and it takes me back to the categories page. The third is on the page when you are viewing other people's suggestions. I clicked that one and it takes me back to where you input their username. All the previous buttons function properly.

I accomplished this in two ways. For the second and third button, I do an on-click function that redirects it to the right page. For the first button, I save the ID's clicked in an array when the previous button is called. I call that array and go to the second from last ID and pass that to my Python where it then pulls all the categories with that parent ID and passes it back, then it is displayed.

User is able to use the tabs at the top to move from page to page.

Before login in I clicked the signup tab and it took me to the register page. I then clicked the login tab and it took me into the login page. I then clicked the home tab and it took me to the home page. From there I logged in and tried the Homepage tab again and it took me there. Then I clicked the My Suggestions tab and it took me to the My Suggestions page with the categories. I then clicked the Their Suggestions tab and it took me to my saved users page to view other people's suggestions.

I accomplished this by using a Navbar in my HTML. I created a frame.html page that connects the tabs to every page. It also only shows home, signup and login in when there is not a user logged in, but when a user is logged in it shows home, my suggestions, their suggestions and logout.

User is able to find someone else's gift suggestions once they have put their username in.

I input another username in the database and clicked the view button.

User is able to use the rating buttons to update the items rating.

I clicked through different thumbs up and down rating buttons. I then went to the database to see if each one I clicked was updated to the appropriate value. I then did the same with the rating buttons for gift suggestions and it was successful as well.

I did this by setting each button with a value. Once the button is clicked it passes that value to my Python. My Python then checks to see if the category or gift already has a rating, if so it updates the rating to the new value. If not, it adds the new rating in.

User is able to click on other's liked categories and gift suggestions.

I logged in as a user and navigated to their suggestions page. I then viewed another user's information. I clicked on different tabs in the gift suggestions to see if it pulled up the associated buttons and it did. I then did the same for the categories tabs and accomplished the same goal.

I did this by using HTML and a used a data-toggle list and role as a tab. This allows each input to be independent and clickable. Once clicked I pull up the Amazon, Walmart, Target, and Google search buttons.

User is able to connect to Amazon with the link provided in other's gift suggestions.

I clicked on a gift suggestion then clicked on the link to Amazon. I was then redirected to an Amazon page that searched the item I clicked on. I did the same process for the category section. Both went to Amazon and searched what I had clicked on.

I accomplished this by doing a simple href in HTML that takes the user to Amazon once clicked. Within the href I took the url from Amazon when you search something and replaced that with my data, the gift suggestion or category that was clicked on.

User is able to connect to Google with the link provided in other's gift suggestions.

I clicked on a gift suggestion then clicked on the link to Google. I was then redirected to a Google page that searched the item I clicked on. I did the same process for the category section. Both went to Google and searched what I had clicked on.

I accomplished this by doing a simple href in HTML that takes the user to Google once clicked. Within the href I took the url from Google when you search something and replaced that with my data, the gift suggestion or category that was clicked on.

User is able to connect to Walmart with the link provided in other's gift suggestions.

I clicked on a gift suggestion then clicked on the link to Walmart. I was then redirected to a Walmart page that searched the item I clicked on. I did the same process for the category section. Both went to Walmart and searched what I had clicked on.

I accomplished this by doing a simple href in HTML that takes the user to Walmart once clicked. Within the href I took the url from Walmart when you search something and replaced that with my data, the gift suggestion or category that was clicked on.

User is able to connect to Target with the link provided in other's gift suggestions.

I clicked on a gift suggestion then clicked on the link to Target. I was then redirected to a Target page that searched the item I clicked on. I did the same process for the category section. Both went to Target and searched what I had clicked on.

I accomplished this by doing a simple href in HTML that takes the user to Target once clicked. Within the href I took the url from Target when you search something and replaced that with my data, the gift suggestion or category that was clicked on.

User is able to use the save button to add the gift suggestions for the categories.

I logged in as a user and navigated to a gift suggestions text box. I put in a gift suggestion and rated it. It was then added to given answers chart. I did this process several times in different categories. All times were successful. I then checked the database to verify that it was added to the database with the proper rating.

I accomplished this by taking in the gift from the text field and checked the items table to see if the current user already had that gift saved; if not, I added the gift and the current user's ID to the table. I then printed it out to the given answers chart. When a user navigates to that page it gathers all the items saved in the items table that is associated with the current user's ID and the category clicked on and outputs the item.

User can save a person's username with their first and last name next to it and go back and look at their gift suggestions later.

I logged in and went to their suggestions tab. I put in another user's username and clicked view. I could see their information. I then hit the previous button to see if the user was saved. The user was added to my saved list with their first and last name next to it. I then put the user's username back into the field and clicked view. I was taken back to the correct page and then hit previous. Upon returning I checked that the user was not added again. I double checked with the database that it added the user and does not add the user twice.

I accomplished this by taking in the username from the text field and searched if the username first exists, if they do, then I checked the others table to see if the current user already has that username saved; if not, I add the username and the current user's ID to the table. I then use that username to gather the information for the page. When the user goes back to the saved users page, it gathers all the users saved in the other table that is associated with the current user's ID and outputs the username, first and last name.

User is able to see the logo and all the homepage info with no problem.

I opened my website and navigated to the homepage. I read through all the different sections and made sure the logo and title lined up correctly and that if the window is not full screen, it resizes properly.

I accomplished my homepage by using HTML with some help from bootstrap to help structure my design.

User is able to use the “my suggestions” button to see their gift suggestions.

I logged in as an already existing user. I then navigated to the My Suggestions button and clicked it. It directed me to the exact page to see the categories and different gift suggestions and their ratings. I double checked this with the database to see if it is giving all the correct information.

I did this by creating a button click function that once clicked will take the user's username and gather their information from the database. It would then be passed back and separated between category and gift suggestions and out put it in the right toggle list.

See if it sets the rating with the highest ratings are at the top when viewing categories and gift suggestions from other users and the user's username.

I created some of my own gift suggestions and rated them differently. I also rated a couple categories. I kept note of what they should be. I then clicked the My Suggestions button to see the results. All the results were printed out exactly how they should. I then double check that it is correct in the database.

I did this by using a sort function that takes an array with all the associated rating, sorts it then outputs it highest to lowest.

I will test if a SQL injection can be performed.

I typed in an SQL injection, `Pizza; SELECT * FROM user`, into all of the available textboxes in the register page, login page, and the text box for gift suggestions. I reviewed the console to see if any information was given, and no information was given either time.

I accomplished this by using SQLAlchemy properly. In SQLAlchemy when you use the add and commit function it eliminates the possibility of a SQL injection

because it is not written in raw SQL and it automatically takes out anything that could be interpreted as a valid SQL command. So to make sure I am fully protected, every time I access the database I used the add and commit functions.

I will check that the passwords are hashed in the database

I went to my database and viewed all the user's password column to make sure that they passwords are hashed.

I did this by using `generate_password_hash`. This hashes the password and over writes the password holding variable so that the password entered cannot be retrieved unless it is in hash form. It is then saved to the database.

I will check that the password being entered is being censored out

I went to the register page and typed in a password. When typing in the password I made sure the characters were being replaced with dots. I did the same

for the login page.

I did this by making that field a password field verses a text field in HTML.

User Test Results

All of the users liked the idea. Some felt I should explain better how the website worked. I added the how it works column on my homepage to better explain this. All of the users did not feel like it was colorful or pulled people in. Had some grammatical errors on the homepage which I fixed. The users thought there could be more categories and sub categories. They would have liked if the rating button stayed highlighted when they go back to the page. One user thought it would be nice if it took you to the Suggestions page as you log in; therefore, upon logging in, a user is redirected to the My Suggestions page. One also suggested putting a button in the My Suggestions page that would take them to user's suggestions. Which I added at the top of the categories. This button is still available when a user clicks to go into a subcategory. This button takes them to exactly what the user sees when they are viewing another user's suggestions, but it is just their information.

Challenges Overcome

What the purpose of the website was

I know what the allotment goal was for the website, I was not sure how to portray it to the users the proper way. My goal was to make gifting easier by answering information about yourself. I went through a couple different ways to do this before I landed on the one I have now. For example, the first way was to ask multiple questions that the user answers. After some consideration, I realized this would be an impossible task without using AI. After a couple more ideas and more failed attempts I came up with the button idea that has different categories. The user is not just answering questions they are saying what they like and do not like as well as adding in some gift suggestions.

Understanding Python

I did not have much knowledge on Python before this project; therefore, I had to overcome the lack of knowledge by doing extensive research on what I was trying to accomplish. I had to learn how to connect my Python to my database

and to my JavaScript. I also learned how to utilize Flask and Flask-Alchemy. I was struggling to understand how to get my users information from my HTML page or how to connect to my database. I had a lot of trial and error to learn how to properly use Flask and Flask-Alchemy.

Understanding JavaScript

Going into this project I had never used JavaScript but once or twice in Applied Networking. I had to learn how to use it with Python and all the functions that I could use to help make the website user friendly and the code simple. I struggled to understand how I could use this to my advantage. I was able to figure out how to connect to my Python and it opened a whole new world. When I started, I did not understand all the possible ways I could use JavaScript.

Understanding Connections

Since I had little experience in most of the languages I used, I had to learn how they could all be connected. This was a big problem for me because I would

see an example online, but I did not understand how to connect the two. It took a lot of research and trial and error to learn how HTML connects to JavaScript, JavaScript connects to Python, and Python connects to SQL database. Without this knowledge I would not have been able to get information from the user and pass it to my database or the other way around.

Future Enhancements

Authorization Request

This is important for security reasons. If this is not implemented, all an attacker needs is the username of who they are going after. Once they have the username, they have the ability to gather information about that person such as first and last name, and their different gift suggestions. Another problem with this is that the user is not able to see who has access to their information. An example of this is a Facebook friend request. It would have a notification bar at the top when a new request has come in. It will show the user's, who sent the request, First and Last name. I will have an accept or decline button. Once the user accepts the other user can see their gift suggestions. If declined they are not given access.

User's friends list

This would be a list of all the people who has access to the user's gift suggestions. In this list you are able to unfriend people. By unfriending them they no longer have access to the user's gift suggestions. There would be a search bar at the top where you can search a username or first and last name of another user. If you add a user, it will then be added to your user list on the "Their Suggestions" tab. This concept is important in that a user is able to stop giving access to someone after the access has been granted.

Vulnerabilities

There are always new and unseen vulnerabilities to look for. I act as a red team and attempt to hack my website to find where my vulnerabilities lie. This is very important in that if I do not protect my users, I would lose their trust and lose users. On top of that losing user information and possibly allowing a hacker to gain access to one of the user's system. All in all, vulnerabilities should be constantly checked for. A way to fix some of the vulnerabilities is to have a setting that will allow the user to decide if everyone, only people shared with, or only they can see their gift suggestions.



More Categories

I would like to add more categories. This is helpful because it will help narrow down what the user likes. This would also help their suggestions because it will be more specific and would help narrow down a gift idea.

Popular Gift Suggestions



I would like to implement when a gift suggestion is used multiple times, it would start to add it as a suggestion to other users. When a user gets to the textbox it would print out the suggested ones, if any, and the user would have the ability to click add or delete. If the user adds it, it will be added to their saved Items list. If they click delete, it will be darkened out, but the user still has the option to add it later. This is helpful because it gives users suggestion, they might not have thought of.

Rating Button



I would code it to where the rating button would stay highlighted when the user goes back to the page. This is helpful for the user's because it allows them to

see what that category is rated from a previous rating. As of now the button is highlighted when on the page but as you move from page to page or login again, it deletes the highlights therefore, the user is unsure what they previously put or if they put anything at all.

Darken Category

To help with the rating button and to make it clearer for the user, I want to darken out a category button when a user clicks the thumbs down. This is helpful for the user to clearly see the categories they like, but it does not completely get rid of the category so if the user changes their mind they would only need to give it a rating of thumbs up to take off the darkening factor.

Design

I would create a better color scheme and logo. I feel what I have now is very dull and lifeless. It does not pull someone to the website. I would have a logo that better represents the website and the colors of the website would coordinate with the new logo. For the pages that are not the homepage, I would add the logo to the

right side of the Navbar as well as the name of the website in the middle. I would also coordinate to the color scheme to the logo.

Error Message

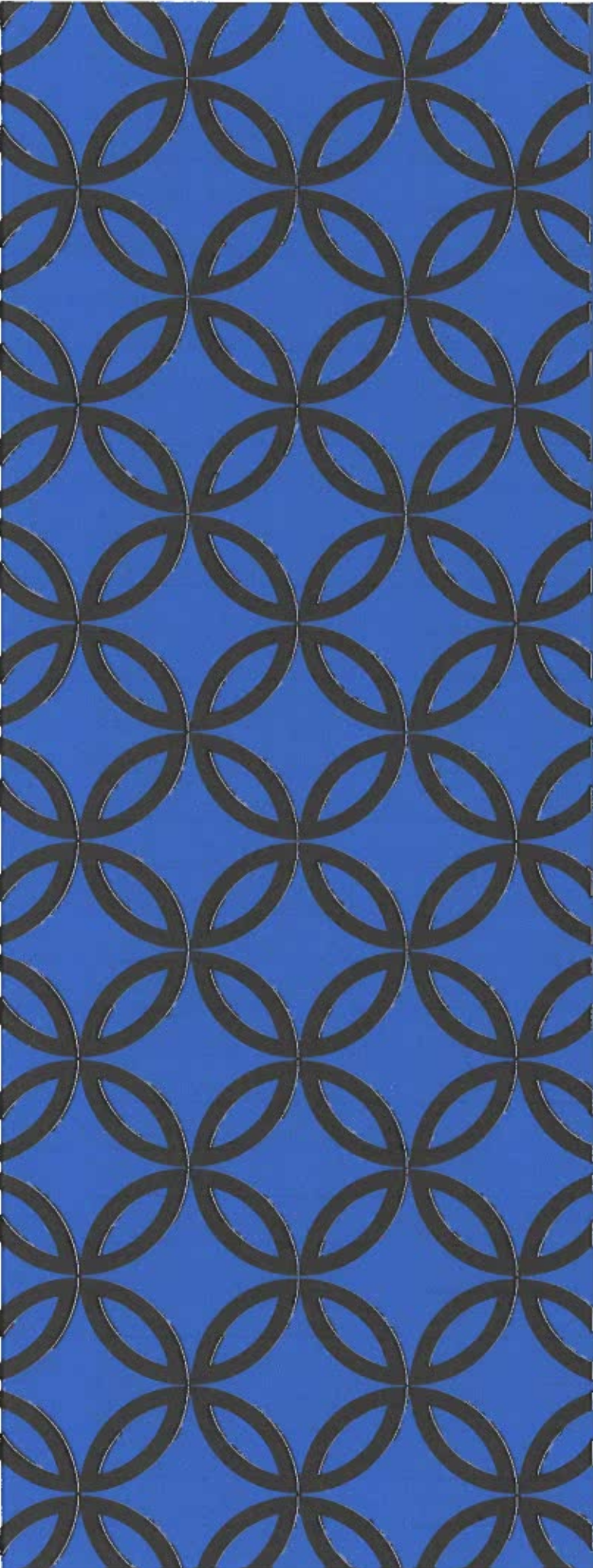
With my criteria for registering and logging in, the error message does not explain exactly what the error was. I would create an error message that would say exactly what criteria the user did not follow. This is helpful for the user because it will tell them exactly what they did wrong, so they are not guessing.

Deploy on Heroku

I would like to deploy my website on Heroku so that it is easily accessible to multiple users. This will allow me to branch out from a local host to a server. This would be very helpful in that users would just need the url and they would have access to the website. Also, the fact that multiple users would be able to use the website at once. I plan to do this with gunicorn which is a helpful tool when trying to deploy a Python app to Heroku.

Address and Thank you

I would also like to implement a way to allow the gift giver a way to mail the package to the user without having to ask for their address. While also allowing the user to mail or email a thank you card to the sender. To do this I would ask for the address when registering and would save it to the user table. I would not implement this until I have the friend request working so that not everyone can see the users address. I would also wait until I get the security settings so that the user can determine who can see their information. I would then have another tab that has user's information such as address. When someone sends a gift it would have a template that the user can use to send a thank you either by mail or email. This is helpful so that the gift giver does not have to ask for their address and the user can send a quick thank you.



GIFTING MADE EASY

B.S. CYBER SECURITY

By: Ashley Ward

STATEMENT OF PURPOSE

- ☐ Have you ever had trouble deciding what to get someone when its Christmas time or for their birthday or for your significant other?
- ☐ If you answered yes then this website is for you!
- ☐ The overall goal of this website is to help choosing a gift for someone easier.
- ☐ It relieves the stress of possible getting something they would not like or something they already have. It also helps eliminate possible duplication of gifts because there are so many more options than just what the person tells you and everyone else who asks.

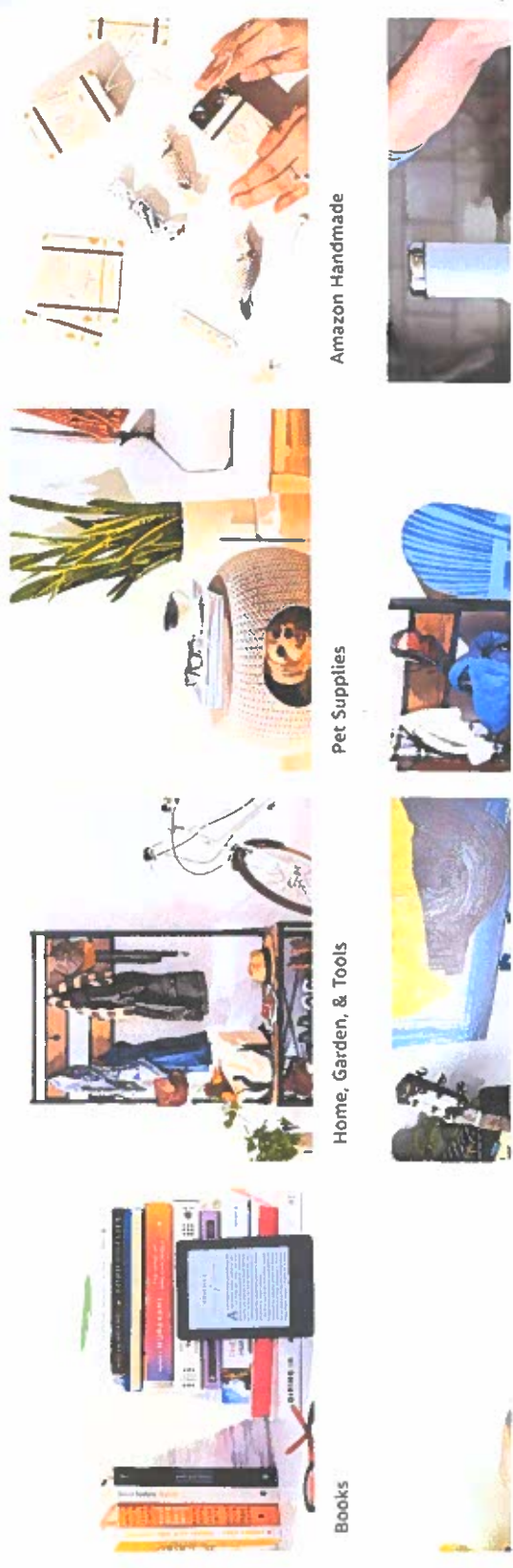
RESEARCH/BACKGROUND

☐ Amazon Gift Registry

RESEARCH/BACKGROUND



Shop by category





RESEARCH/BACKGROUND CONTINUED

- ☐ Python
 - ☐ What is it?
 - ☐ How I used it.
- ☐ HTML, CSS
 - ☐ What is it?
 - ☐ How I used it.
- ☐ JavaScript
 - ☐ What is it?
 - ☐ How I used it.

RESEARCH/BACKGROUND

☐ SQLAlchemy

- ☐ What is it?
- ☐ How I used it.

☐ Flask

- ☐ What is it?
- ☐ How I used it.

☐ Flask-Alchemy

- ☐ What is it?
- ☐ How I used it.

SOFTWARE

- ☐ Software
 - ☐ Visual Studio Code
 - ☐ Flask
 - ☐ Python
 - ☐ HTML
 - ☐ CSS
 - ☐ JavaScript
 - ☐ Bootstrap
 - ☐ SQL
 - ☐ SQLAlchemy
 - ☐ Flask-Alchemy
 - ☐ Virtual Environment

HANDS ON WALK THROUGH

TEST PLAN

- ☐ Possible SQL injections
- ☐ Hashing Passwords
- ☐ I tested the user side.
 - ☐ Signing up
 - ☐ Logging in
 - ☐ Navigating from page to page
 - ☐ Navigating through the categories
 - ☐ Rating different categories
 - ☐ Adding gift suggestions and rating them
 - ☐ Lookup another user and viewing their gift suggestions
 - ☐ Using the links to search for their gifts
 - ☐ Logging out

TEST RESULTS

- ☐ In the end all of my tests passed
- ☐ I had several people test my website as a user and gathered their feedback
 - ☐ All liked the idea and the purpose of it
 - ☐ Got several suggestions to improve the website as well as some grammatical fixes

CHALLENGES OVERCOME

- ☐ What the purpose of the website was
- ☐ Understanding Python
- ☐ Understanding JavaScript
- ☐ Understanding the different connections

FUTURE ENHANCEMENTS

- ☐ Authorization Request like a friend request
- ☐ User's friends list to list out friends and can delete them
- ☐ Other possible vulnerabilities
- ☐ More categories and subcategories
- ☐ Popular gift suggestions
- ☐ Rating button always highlighted
- ☐ Darken Category if thumbs down
- ☐ Design of website such as logo and colors
- ☐ Error message saying what the user types in incorrectly for login and sign-up
- ☐ Deploy on Heroku